

$$\text{lex } x^2$$
$$\text{wolf}$$

EE45: Intro to Foundations of ML

Mod 1 - Lec 1: Data Fitting Concepts

Course Overview Reminder

- Focus on
 - ▶ **simple, core models**: e.g., linear regression & classification),
 - ▶ **fit models via optimization**: gives us a framework for learning from data, generalization, and interpretability
 - ▶ and the **mathematics of ML**: enables us to understand (transparently) and improve models

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 - ▶ **fit models via optimization**: gives us a framework for learning from data, generalization, and interpretability
 - ▶ and the **mathematics of ML**: enables us to understand (transparently) and improve models
- Show connections to applications in
 - ▶ data science & statistics,
 - ▶ signal processing,
 - ▶ decision-making & control,
 - ▶ computer vision/perception,
 - ▶ AI and robotics

Overview: [Module 1] Data Fitting Concepts

- This week
 - ▶ How to measure "fit" of a model to data?: norms, distances, inner products
 - ▶ Example: k -means clustering

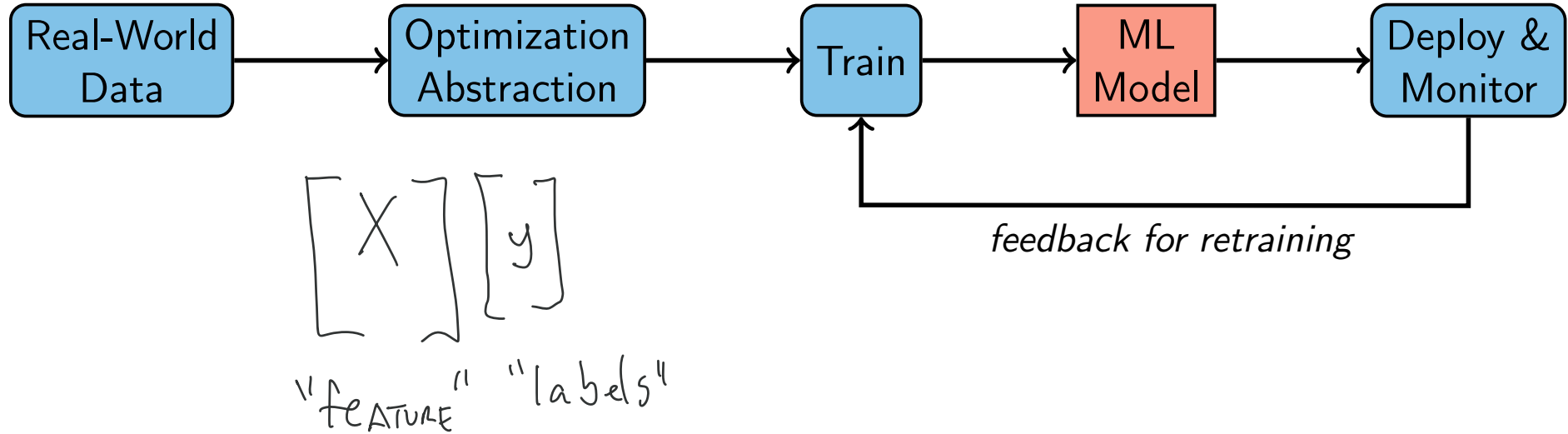
References:

- [VMLS]: Chapters 1–3
- Topics: Vectors, Inner product, Linear functions, Regression model, Norm and distance

Intro to Applied LA: Vectors, Matrices, & Least Squares

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ML workflow: Role of Linear Algebra & Optimization



This course: Show that many interesting learning and optimization problems are framed and solved using language and techniques from **Linear Algebra** and **Optimization**

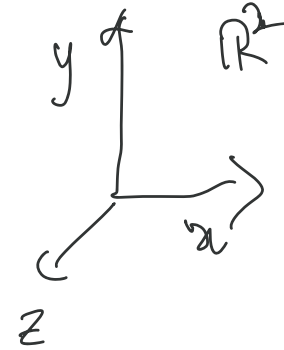
ML workflow: Role of Linear Algebra & Optimization

Real world task involving data $\longrightarrow$ Optimization Problem $\longrightarrow$ ML model

Canonical Optimization Problem Terminology:

$$\min \ell(\theta)$$

$$\theta \in \mathcal{H} \subseteq \mathbb{R}^n$$



- $\ell(\theta)$: loss fn.
- θ : model parameters
- $\mathcal{H} \subseteq \mathbb{R}^n$ "constraint set"

$$\theta = (\theta_1, \theta_2, \dots, \theta_n)$$



$$|\theta_1| \leq c$$

Formulating a Learning Problem as Optimization

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- Machines need to measure similarity, and distances are the key
 - ▶ Norms and distances turn subjective ideas (are these two cats alike?) into math: $\|x - y\|$.
 - ▶ Different norms capture different ideas of similarity (Euclidean vs Manhattan vs cosine).

$$\theta = (\theta_1, \theta_2) \rightarrow \sqrt{\theta_1^2 + \theta_2^2}$$

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Model

A model is a recipe or rule that takes input data and produces an output (a prediction).

• $f_\theta: \mathcal{X} \rightarrow \mathcal{Y}$ that takes $x \in \mathcal{X}$ produce output
 $\mathbb{R}^m$ $\mathbb{R}$ "live in" $\hat{y} = f_\theta(x) \in \mathcal{Y}$

EXP: $\mathcal{X} \subseteq \mathbb{R}^m$, $x = (x_1, x_2, \dots, x_m)$
 $m=2$, $x = (x_1, x_2)$

Model fitting as optimization

$$f_{\theta}(x) = \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 + \theta_4$$

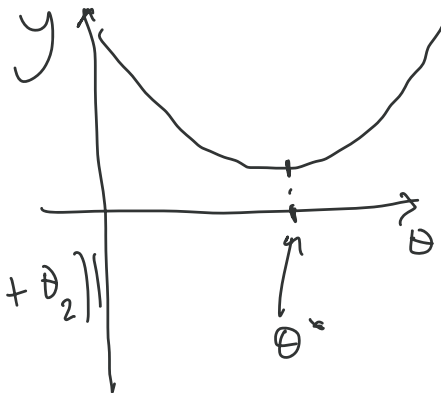
Goal

Given data $\{(x_i, y_i)\}_{i=1}^n$ and a model f_{θ} , choose parameters θ to best explain the data.

$$\theta^* = \underset{\theta \in \Theta}{\operatorname{argmin}} \mathcal{L}(\theta)$$

$$\min_{\theta \in \Theta} \mathcal{L}(\theta)$$

$$\begin{aligned} \ell(y_i, f_{\theta}(x_i)) &= |y_i - f_{\theta}(x_i)| \\ &= |\text{temp in room}(i) - (\theta_1 (\text{# people}) + \theta_2)| \end{aligned}$$



$$\mathcal{L}(\theta) = \frac{1}{n} \sum_{i=1}^n \ell(y_i, f_{\theta}(x_i))$$

$$x_i = \begin{bmatrix} \text{\# people in room} \\ \text{vol air flow} \\ \text{temp outside} \end{bmatrix}, y_i = [\text{temp in room}]$$

Key idea: *measuring error* $\longleftrightarrow$ *optimization objective*

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Then we **optimize** that number:

$$\theta^* = \arg \min_{\theta} \frac{1}{n} \sum_{i=1}^n \ell(y_i, f_{\theta}(x_i))$$

Eg. LEAST SQUARES

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Design choice: the loss ℓ is usually built from a **distance** (how far?) or a **similarity** (how aligned?).

Formulating Optimization: Similarity & "Distance" Measures

- Similarity is the hidden engine behind search, recommendation, and personalization.
 - ▶ Netflix recommends shows based on similarity to things you've watched.
 - ▶ Google finds web pages since they're similar in content to your query.
 - ▶ Spotify suggests songs since your playlist is similar to other users' playlists.

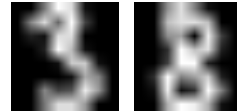
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- Fundamental ML Operations Depend on It
 - ▶ **Classification:** Assign a label to a new point based on similar past examples (nearest neighbors).
 - ▶ **Clustering:** Group data points that are similar (k-means).
 - ▶ **Dimensionality reduction:** Keep points that are similar in the original space close together in a compressed representation (PCA, t-SNE).
 - ▶ **Representation learning (modern ML):** Train embeddings so that similar meanings (words, images, faces) are close in vector space.

Similarity

- How do we decide if two people, two images, or two signals are similar?
- examples: Two histories of Spotify users', two patients' health metrics, etc.
 - ▶ Two grayscale digit images of "3" and "8".

8x8 images

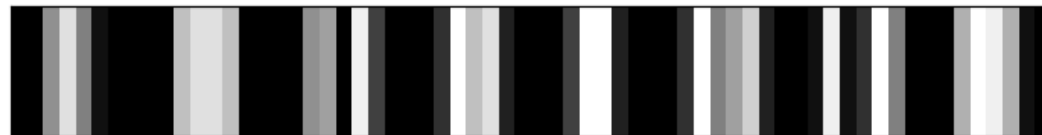


- represent each object as a vector in some space. Similarity is then about distance between vectors.

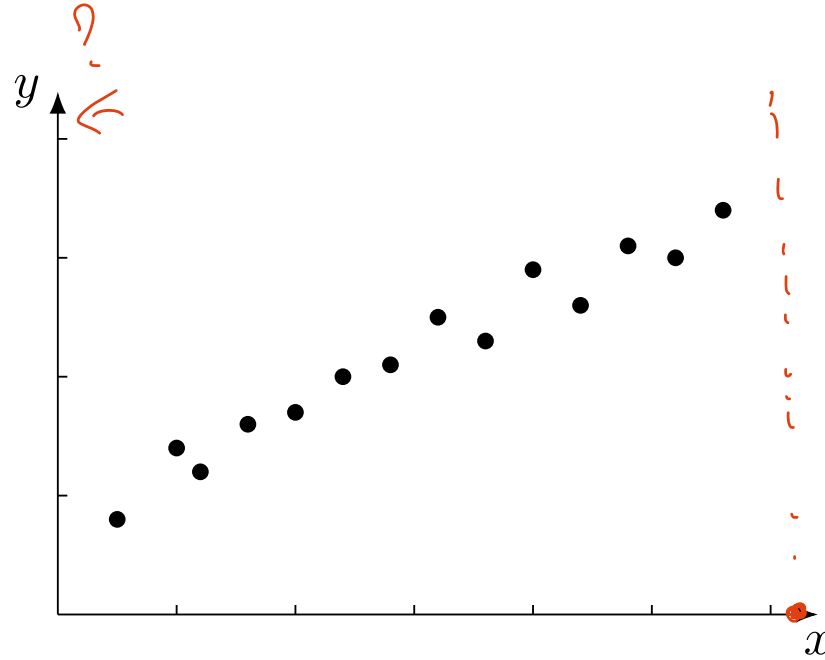
Digit 3 Vector Strip



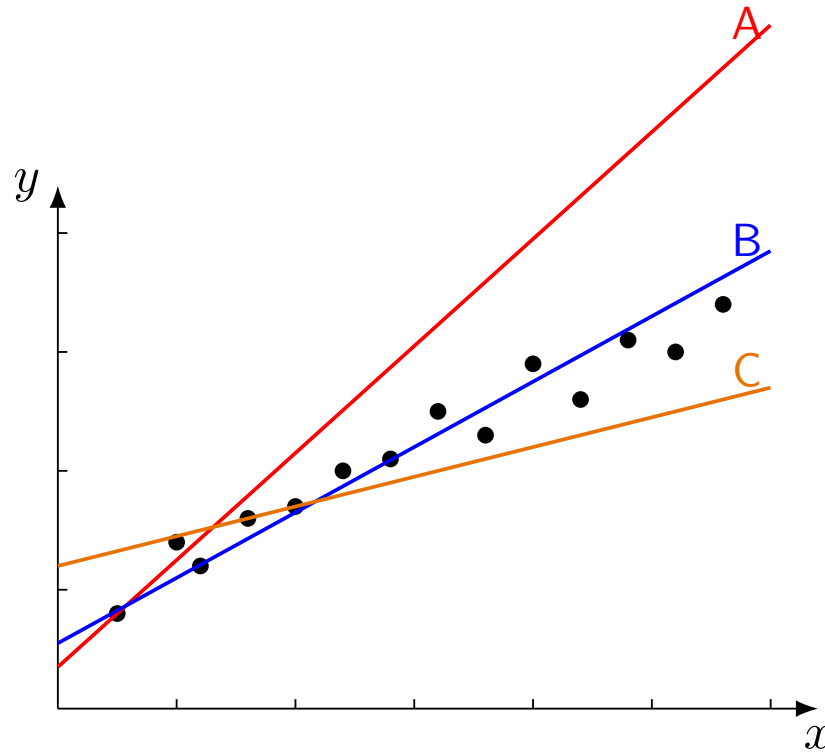
Digit 8 Vector Strip



Model "Fit": Which line fits these points best?

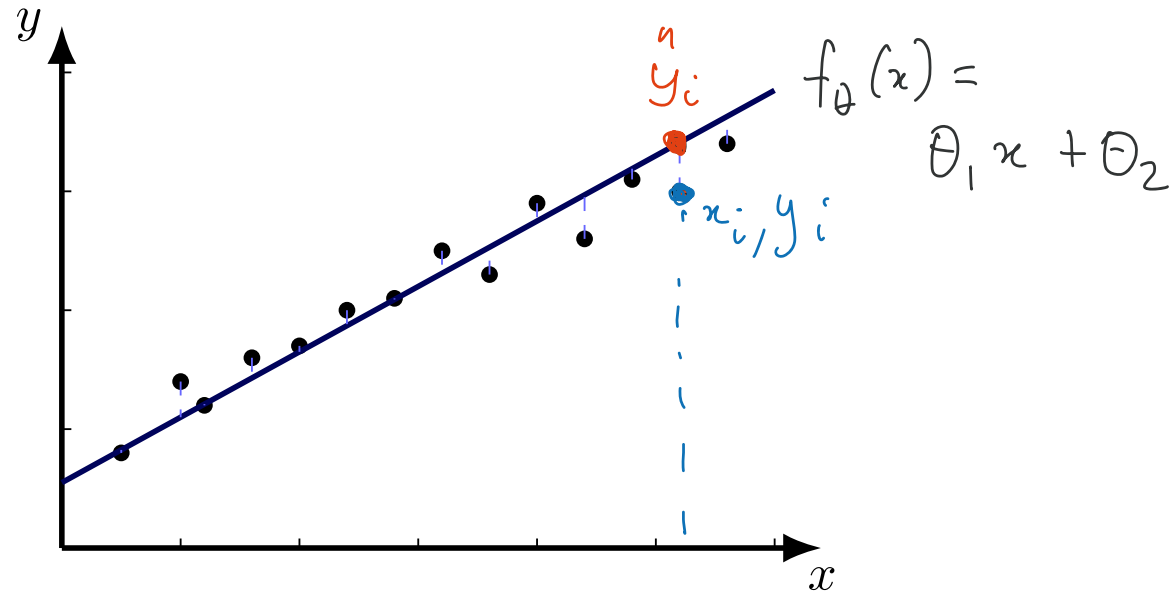


Model "Fit": Which line fits these points best?



Which line would you pick, and why?
How can we make this decision mathematically?

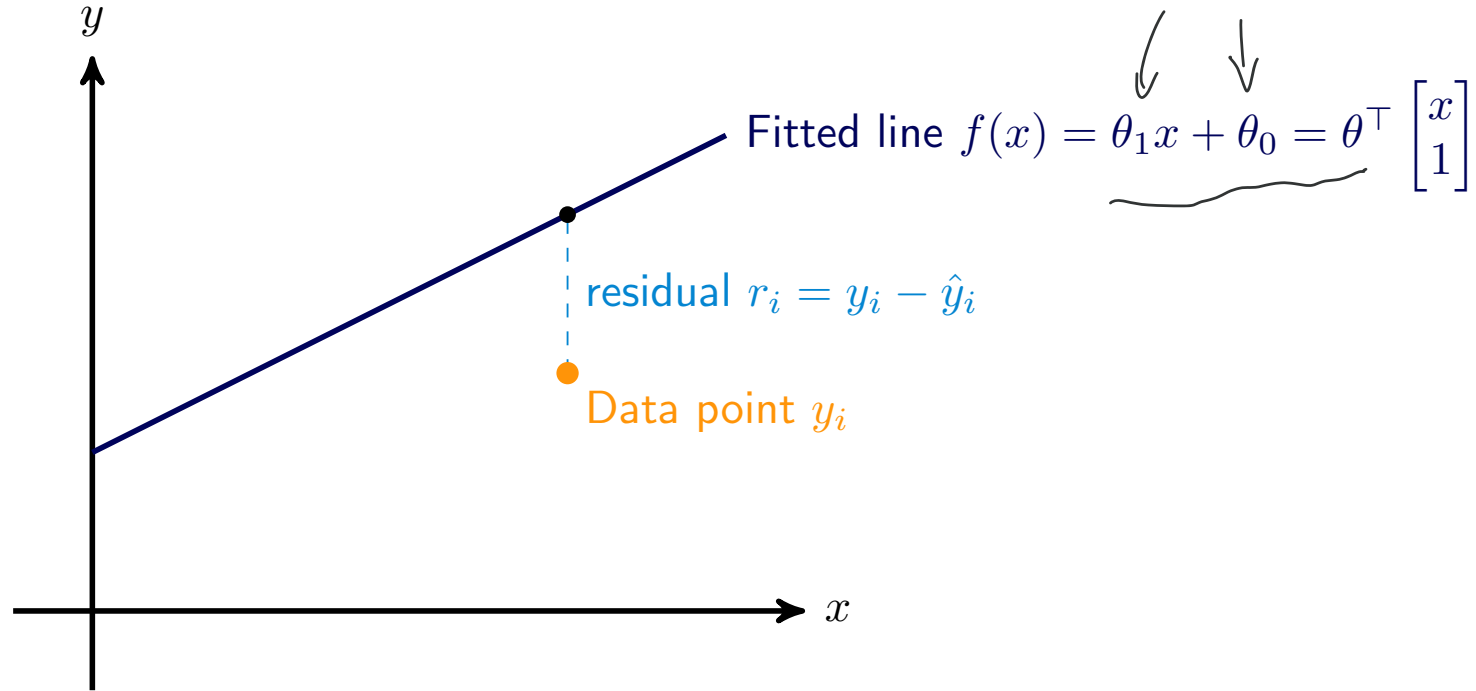
Residuals: distance from points to a line



Residuals (vertical distances) to candidate B.
Least squares picks the line that minimizes the sum of squared residuals.

$$r_i = y_i - \hat{y}_i$$

Residuals



$$\|y - X\theta\|_2$$

Least Squares Vector Notation

$$\theta \in \mathbb{R}^2$$

different metric $\rightarrow$

$$|y_i - (\theta_1 x_i + \theta_0)|$$

This problem is easily written in vector notation.

$$\min_{\theta \in \mathbb{R}^2} \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \min_{\theta \in \mathbb{R}^2} \sum_{i=1}^n (y_i - (\theta_1 x_i + \theta_0))^2$$

where

- $y = (y_1, \dots, y_n)$

$$\Rightarrow \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$$

$$X = \begin{bmatrix} x_1 & 1 \\ x_2 & 1 \\ \vdots & \vdots \\ x_n & 1 \end{bmatrix} \quad (n \times 2)$$

$$\theta = \begin{bmatrix} \theta_1 \\ \theta_0 \end{bmatrix} \quad 2 \times 1$$

$$y - X\theta = \begin{bmatrix} y_1 - (\theta_1 x_1 + \theta_0) \\ y_2 - (\theta_1 x_2 + \theta_0) \\ \vdots \\ y_n - (\theta_1 x_n + \theta_0) \end{bmatrix} \Rightarrow \|y - X\theta\|_2^2$$

top

Vectors: notation

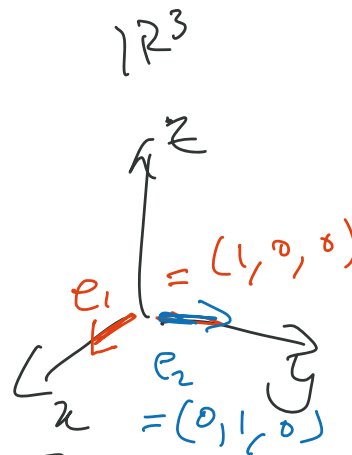
- a vector is an ordered list of numbers: $(-1.2, 0, 3.6)$, or

$$\begin{bmatrix} \end{bmatrix}^T \begin{bmatrix} \end{bmatrix}$$

~~n x 1~~ $m \times 1$
 $1 \times m$

$$x = \begin{bmatrix} -1.2 \\ 0 \\ 3.6 \end{bmatrix} \in \mathbb{R}^3$$

- $x \in \mathbb{R}^m$ $(x_1, \dots, x_m)$ x_i entry Real #



$$x^T \quad m \quad \begin{bmatrix} x_1 \\ \vdots \\ x_m \end{bmatrix}^T \rightarrow [x_1 \dots x_m] \quad \begin{matrix} z=m \\ \begin{bmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & \dots & \dots \\ x_{31} & \dots & \dots \end{bmatrix}^T \end{matrix} \rightarrow \begin{matrix} n=2 \\ \begin{bmatrix} x_{11} & x_{21} & x_{31} \\ x_{12} & x_{22} & x_{32} \end{bmatrix} \end{matrix} \quad \begin{matrix} m=3 \end{matrix}$$

$$\underline{\underline{1}}_m = (1, 1, \dots, 1) \quad , \quad e_i = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \leftarrow i^{th} \quad e \quad \mathbb{R}^3$$

Examples

- (x_1, x_2) represents a location or a displacement in 2D
- examples:
 - ▶ portfolio: entries give shares (or \$ value or fraction) held in each of n assets, with negative meaning short positions
 - ▶ cash flow: x_i is payment in period i to us
 - ▶ sampled audio signal: x_i is the acoustic pressure at sample time i
 - ▶ feature vector: x_i is the value of i th feature or attribute of an entity
 - ▶ word count vector

BAG of words
Lang. models (pre-Neural)
NN Lang. models (word # $\rightarrow$ feature)

Word count example

- a short document:

“Word count vectors are used in computer based document analysis. Each entry of the word count vector is the number of times the associated dictionary word appears in the document.”

- dictionary & word count vector:

word

in

number

flower

the

document

$\begin{bmatrix} 3 \\ 2 \\ 1 \\ 0 \\ \vdots \end{bmatrix}$